Angles from intersecting chords, secants, and tangents

Core Concepts: Angles from intersecting chords, secants, and tangents

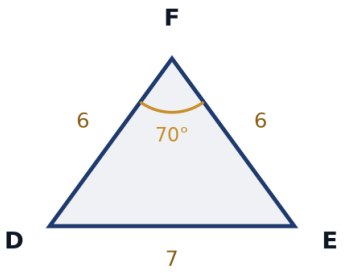
1 In $\triangle ABC$ studying angles from intersecting chords, secants, and tangents, $m\angle A = 43^\circ$ and $m\angle B = 45^\circ$. What is $m\angle C$?

a. 92°

b. 102°

c. 84°

d. 137°



2 A geometric figure scaled by factor $k = 2$ has an original side length of 7 cm. What is the corresponding image length?

a. 12 cm

b. 14 cm

c. 21 cm

d. 28 cm

3 On the coordinate plane, a line with slope $m = 5$ passes through $(0, 3)$. What is its y-coordinate when $x = 1$?

a. 3

b. 10

c. 8

d. 13

Procedural Fluency & Algebraic Manipulation

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4 In $\triangle ABC$ studying angles from intersecting chords, secants, and tangents, $m\angle A = 43^\circ$ and $m\angle B = 52^\circ$. What is $m\angle C$?

a. 85°

b. 95°

c. 77°

d. 137°

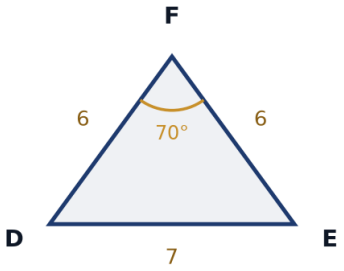
5 A geometric figure scaled by factor $k = 3$ has an original side length of 7 cm. What is the corresponding image length?

a. 18 cm

b. 21 cm

c. 28 cm

d. 63 cm



6 On the coordinate plane, a line with slope $m = 5$ passes through $(0, 3)$. What is its y-coordinate when $x = 2$?

a. 8

b. 15

c. 13

d. 18

Graphical & Geometric Interpretation

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7 In $\triangle ABC$ studying angles from intersecting chords, secants, and tangents, $m\angle A = 43^\circ$ and $m\angle B = 59^\circ$. What is $m\angle C$?

a. 78°

b. 88°

c. 70°

d. 137°

8 A geometric figure scaled by factor $k = 4$ has an original side length of 7 cm. What is the corresponding image length?

a. 24 cm

b. 28 cm

c. 35 cm

d. 112 cm

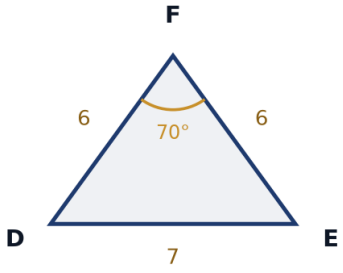
9 On the coordinate plane, a line with slope $m = 5$ passes through $(0, 3)$. What is its y-coordinate when $x = 3$?

a. 13

b. 20

c. 18

d. 23



Exam-Style Problem Solving & Modeling

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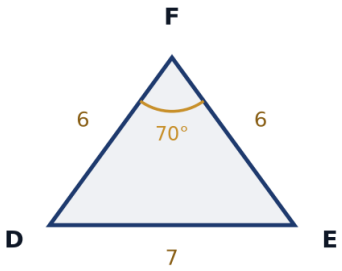
10 In $\triangle ABC$ studying angles from intersecting chords, secants, and tangents, $m\angle A = 43^\circ$ and $m\angle B = 66^\circ$. What is $m\angle C$?

a. 71°

b. 81°

c. 63°

d. 137°



11 A geometric figure scaled by factor $k = 5$ has an original side length of 7 cm. What is the corresponding image length?

a. 30 cm

b. 35 cm

c. 42 cm

d. 175 cm

12 On the coordinate plane, a line with slope $m = 5$ passes through $(0, 3)$. What is its y-coordinate when $x = 4$?

a. 18

b. 25

c. 23

d. 28

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